

Replicating the Tsukorok receiver chain

Hardware and development boards (receive-only replication; AERIX RF policy excludes any transmit use)

1. What the original uses (CONFIRMED from firmware)

Block	Part	Bus / pins in firmware	Notes
MCU	ESP32-S3 (Xtensa LX7), Arduino-ESP32 / ESP-IDF 4.4.7	-	Wi-Fi, OLED, buzzer, SPIFFS captures, CLI
2.4 GHz	Semtech SX1280 (RadioLib SX128x)	SPI SCK17 MISO15 MOSI16 NSS18, IRQ14 RST12 BUSY13, 2 MHz	Used in FSK mode 250 kbps/195 kHz for RSSI sweeps, LoRa 812.5 kHz for the ELRS scan
Sub-GHz	Semtech SX1276/SX1278 (RadioLib SX127x)	SPI SCK7 MISO6 MOSI8 NSS9, 10 MHz, DIO0 ISR	FSK 15-80 kbps, 76.7 kHz dev, 200 kHz RX BW; LoRa 500 kHz CAD; 755 MHz init, sweeps 860-1020 MHz
5.8 GHz	AMICCOM A5133 (bit-banged 3-wire SPI)	SDIO16 SCK17 CS48 (shares 16/17 with the SX1280 bus)	5725-5899 MHz in 1 MHz steps, 8-bit RSSI. Rarely stocked outside Asia
2.4 GHz alt.	TI CC2500	present, "cc2500 unsupported" for scans	legacy path, not needed
Power	GPIO21 RF section enable	-	radios powered only during a tick

2. Development boards that replicate it

Everything below is receive-capable with RadioLib on ESP32-S3; transmit functions are simply never called. Prices are indicative.

Option	Boards / modules	Covers	Comment
Closest one-board match	LilyGO T3-S3 (ESP32-S3 + radio + OLED), available with SX1280 (2.4 GHz LoRa) or SX1276/SX1262 (sub-GHz)	2.4 GHz OR sub-GHz per board	Two T3-S3 boards (one SX1280, one SX1276) reproduce the SX-radio parts almost pin-for-pin in spirit; RadioLib examples exist for both.
Modular build	ESP32-S3-DevKitC-1 (N8R8) + Ebyte E28-2G4M12S/E28-2G4M27S (SX1280) + HopeRF RFM95W/RFM96W or Ebyte E19-868M30S (SX1276)	2.4 GHz + 860-1020 MHz	Choose the 868/915 MHz SX1276 variant (RFM95/96), not the 433 MHz SX1278 modules (Ra-01/02) - the sweeps go to 1020 MHz. SX1276 tunes 137-1020 MHz.
Heltec alternative	Heltec WiFi LoRa 32 V3 / LilyGO T-Beam (SX1262)	sub-GHz	SX1262 is not the same chip; RadioLib supports it, but the CAD/receive-fast register tricks of the firmware are SX127x-specific and would need re-implementation.
5.8 GHz, faithful	AMICCOM A5133 module (vendor modules from Taiwan/China; part is 5.8 GHz 15 dBm FSK, 3/4-wire SPI)	5725-5899 MHz	Hard to source; and note the firmware's TX-strobe misuse - a replication must use Standby/PLL strobes (0xA0/0xB0) to leave RX.
5.8 GHz, practical	RX5808 / RTC6715 analog video receiver module with RSSI pin, SPI-tunable (5645-5945 MHz)	5.8 GHz RSSI sweeps	Receive-only by design, cheap, widely used with ESP32 in FPV diversity projects. Reproduces D1/D2 style sweeps (RSSI vs channel), not FSK reception.
Full-band SDR	HackRF One (1 MHz-6 GHz, 20 MS/s) - already the AERIX RF secondary backend	all three bands	Reproduces every sweep as a spectrum, and adds what the device lacks: IQ, syncword correlation, timing. Not real-time for the packet-level FSK receive unless a demodulator is implemented.
Primary platform	ANTSDR E200 (AD9361, 70 MHz-6 GHz, 2x2)	all three bands, coherent	The only option here that supports TDOA-grade timing (with GPSDO/reference). Use it to validate the syncwords and to measure the Tsukorok's own 5.8 GHz emissions.

3. Recommended replication plan

- **Step 1 - sub-GHz FSK/LoRa (highest-value methods A1/A2/B1):** ESP32-S3 + SX1276 module (RFM95W or E19-868M30S), RadioLib, beginFSK(755 MHz, br/1000, 76.7 kHz, 200 kHz, 10 dBm, preamble 16) then sweep

with setFrequency/setBitRate exactly as the firmware does; log the first 4 bytes after preamble. Compare against the nine static words.

- **Step 2 - 2.4 GHz spectral:** second board or a T3-S3 SX1280; FSK 250 kbps, 400 kHz steps, GetRssiInst. Implement the 16-window and the score-A/B features from the brief; expect the same false alarms on Wi-Fi.
- **Step 3 - 5.8 GHz:** RX5808 with RSSI to ADC for the sweep-based rules; skip the A5133 unless a module is available, and never use TX strobes.
- **Step 4 - cross-check with the SDRs:** capture the same bands with HackRF/E200 while the replica runs; this is what upgrades vendor syncwords from level 2 to level 3 evidence and what a TDOA experiment needs anyway.

Cost indication: two T3-S3 boards plus an RX5808 module is under 100 EUR; the SDR chain already exists in the project.